# Alex Spaulding

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## Professional Summary

Computer Science student building systems software, high-performance applications, and AI-integrated tools. Strong foundation in C, Rust, and TypeScript with a focus on systems design, low-level optimization, and AI integration. Seeking Software Engineering internship opportunities.

## Education

**Eastern Washington University** — *Spokane, WA*

*Bachelor of Science (BS) in Computer Science, Math Minor* — Feb 2025 - Present

- **Relevant Coursework**: Algorithms, Computer Architecture, C & UNIX Programming, Cybersecurity, Relational Databases, Computer Networks

**The University of Montana** — *Missoula, MT*

*Some College* — Sep 2021 - 2024

- **Relevant Coursework**: Data Structures, Software Principles I & II, Operating Systems & Command Line Interface

## Technical Skills

**Languages**: C, C++, Rust, Swift, Python, Java, JavaScript/TypeScript, SQL, Bash, Nix
**Systems & Security**: Unix, Inter-Process Communication (IPC), macOS/iOS internals, Reverse Engineering, Mach-O, Kernel/Runtime Mods, Git, Continuous Integration/Deployment (CI/CD)
**AI Integration**: OpenAI APIs, Large Language Model (LLM) integration, Whisper, GPT, Text-to-Speech synthesis
**Frontend & Backend**: React, Next.js, Node.js, REST APIs, Docker, Amazon Web Services (AWS), Google Cloud Platform (GCP), HTML/CSS

## Projects

**apple-sharpener (269 ★)** — *GitHub*

*macOS System Customization (Objective-C, C)* — 2023 - Present

- Built a system-level macOS customization tool earning **269 GitHub stars**; resolved **15+ critical runtime bugs** through low-level performance profiling.
- Implemented Objective-C runtime hooks and Mach-O patching to optimize system behaviors and UI responsiveness.

**Wawona (96 ★)** — *Wawona.io* / *GitHub*

*Native Wayland Compositor (Rust, C, Nix)* — 2025 - Present

- Engineered a cross-platform Wayland compositor enabling **native Linux app execution** on macOS/iOS/Android; optimized rendering for low-latency performance.
- Designed reproducible build pipelines using Nix and automated CI/CD verification for cross-architecture deployments.

**Whisperer** — *GitHub*

*AI Voice Assistant for watchOS (Swift, Node.js)* — 2023 - Present

- Developed a voice-powered ChatGPT client for Apple Watch; integrated OpenAI Whisper for transcription and Vercel Serverless Functions for real-time math rendering.

## Experience

**University of Montana IT Help Desk** — *Missoula, MT*

*IT Consultant* — Sep 2020 - Jan 2025

- Resolved Tier I technical issues for **20+ users weekly** and managed **1,200+ service tickets** via TeamDynamix.
- Diagnosed complex network, hardware, and account security failures across campus-wide systems.

**Sunburst Sensors** — *Missoula, MT*

*Assembly Technician* — Jan 2022 - Mar 2025

- Assembled pCO2/pH instruments; performed PCB soldering, precision fiber optic cleaving, epoxy potting, and machining.
- Executed bulk firmware updates for **1,000+ microcontrollers**; utilized proprietary software for sensor calibration and optimization.
- Fabricated brass/optical filters and performed hydrostatic pressure testing; spot welded **300+ custom battery packs**.

**Wheelbound Productions** — *Missoula, MT*

*Web Developer* — Apr 2018 - Sep 2018

- Launched **700+ responsive web pages** for dumonds.com utilizing mobile-first architectures; optimized performance for high-traffic production sites.

## Leadership & Awards

**VP – Association for Computing Machinery (ACM)** – Led technical events, Hackathons, and Capture The Flag (CTF) competitions for the student community — 2025 - Present

**1st Place – Gonzaga University Hackathon** – DevOps Lead; architected CI/CD pipelines for real-time game engine — 2025

**1st Place – UM Cyber Competition** – CTF winner; performed network reconnaissance and exploitation — 2024

**Front End Certificate** – 18-week program; 18 projects including a (Final Project) — Jul 2021

*References available upon request.*